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Simulation and Sensor Modeling

APS

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CHECKPOINT

ADAPTIVE PATHOGEN SURVEILLANCE | INDIVIDUAL PROGRESS BRIEF

MY ROLE

I focus on the simulated facility and how its sensors behave. My part is to make the readings understandable and repeatable so the rest of our team can test source-location methods.

OUR WORKING FOUNDATION

Our shared prototype represents eight connected zones, chooses an outbreak origin that the prediction system does not know, and generates readings that can include noise or missing sensors. I am building on that working simulation, not claiming that it models a real pathogen outbreak.

I am treating this as a controlled classroom model rather than a detailed airflow or disease-transmission model.

MY CHECKPOINT STATUS

I am: [building / testing / refining]

MY CONTRIBUTION SO FAR

I have completed: [specific file, test, or output]

TERMS AND IDEAS I USE

I use NetworkX, a Python library for connected networks: each facility zone is a node and each connection is an edge. The hidden source is the chosen origin of a simulated event. Graph distance counts the connections along the shortest route between zones. Signal decay means readings generally weaken farther from the source. A random seed lets me repeat a run, while noise changes readings and missing data leaves some unavailable.

WHAT I AM WORKING ON NEXT

1. I will rerun several source locations with fixed random seeds to confirm the same settings produce the same results.
2. I will compare ideal readings with noisy or missing readings and check that the distance-based pattern makes sense.
3. I will document the true source, per-zone observations, and simulator settings in a format the data and prediction components can use.

CHALLENGES I AM CONSIDERING

The shortest route through connected zones is not the same thing as actual airflow. I need to explain that limitation and record settings that produce unexpected results.

WHAT I PLAN TO SHOW

I plan to show one repeatable simulation, the relevant code or test, and the assumptions behind its sensor readings.

HOW I WOULD EXPLAIN MY WORK

My checkpoint summary: I am working on the facility simulation and sensor behavior. The simulator lets us start with a known origin, hide that answer from the prediction method, and see whether imperfect readings still contain useful location information. My immediate focus is repeatability and clear assumptions.